

B.Tech. Degree IV Semester Regular/Supplementary Examination June 2023

ME 19-205-0405 HYDRAULIC MACHINERY
(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

CO1: Understand dimensional analysis and principles of similitude.

CO2: Apply the momentum principles to impinging jets and analyse its performance.

CO3: Learn the working of various hydraulic turbines and their performance characteristics.

CO4: Understand the working of roto-dynamic pumps and positive displacement pumps.

CO5: Study the principle of working of hydraulic devices.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A

(Answer ALL questions)

	(8 × 3 = 24)	Marks	BL	CO	PO
I. (a) Compare the Rayleigh's method and Buckingham's Pi theorem method of dimensional analysis.	3	L1	1	1	
(b) Starting from fundamentals obtain the maximum efficiency of an independently moving flat plate with a velocity 'u' hit by a jet moving with a velocity of 'v'.	3	L2	2	1	
(c) Classify hydraulic turbines based on their working principles. Give examples for each.	3	L1	3	1	
(d) Explain draft tube theory for turbines.	3	L1	3	1	
(e) Explain minimum starting speed of a centrifugal pump.	3	L2	4	1	
(f) Draw an indicator diagram of a reciprocating pump. Prove that area of indicator diagram is proportional to the work done by the reciprocating pump.	3	L2	4	1	
(g) Explain hydraulic lift with neat sketch.	3	L2	5	1	
(h) Compare hydraulic intensifier and hydraulic accumulator.	3	L1	5	1	

PART B

(4 × 12 = 48)

II. (a)	Apply Buckingham's Pi theorem to express η in terms of dimensionless parameters. Where η the efficiency of a fan depends on density ρ , dynamic viscosity μ of the fluid, angular velocity ω , diameter D of the rotor and discharge Q .	8	L3	1	1
(b)	A pipe of diameter 1.5 m is required to transport an oil of specific gravity 0.9 and viscosity 3×10^{-2} poise at the rate of 3000 litre/s. Test were conducted on a 15 cm diameter pipe using water at 20°C. Find the velocity and rate of flow in the model. Viscosity of water at 20°C = 0.01 poise.	4	L3	1	1
OR					
III. (a)	Obtain the equation for the work done per second by a jet of water striking at the centre of an independently moving symmetrical curved vane.	4	L3	2	1
(b)	A jet of water having a velocity of 35 m/s impinges on a series of vanes moving with a velocity of 20 m/s. The jet makes an angle of 30° to the direction of motion of the vanes when entering and leaves at an angle of 120° . Draw the triangles of velocities at inlet and outlet and find: (i) the angles of the vane tips so that water enters and leaves without shock (ii) the work done per unit weight of water entering the vanes (iii) the efficiency.	8	L3	2	1

(P.T.O.)

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		Marks	BL	CO	PO
IV.	(a) Design a suitable hydraulic turbine for the following power plant. In a hydro electric power plant the turbine is to develop 1 MW at a speed of 1000 rpm. The survey reveals that 450 m of static head and a minimum flow of $0.3 \text{ m}^3/\text{s}$ is available. Find the specific speed of the turbine. Assume overall efficiency as 80%. Assume other ratios within permissible limits.	8	L3	3	2
	(b) Explain governing of Pelton turbines.	4	L2	3	1
OR					
V.	(a) Design a Francis turbine runner with the following data. Net head $H = 68\text{m}$, Speed $N = 750 \text{ rpm}$, output power $P = 330 \text{ KW}$, $\eta_h = 94\%$, $\eta_o = 85\%$, flow ratio = 0.15, breadth ratio = 0.1, inner diameter of the runner is $\frac{1}{2}$ of the outer diameter. Also assume 6% of the circumferential area of the runner to be occupied by the thickness of the vanes. Velocity of flow remains constant throughout and flow is radial at exit.	8	L3	3	2
	(b) Explain main characteristic curves of a Kaplan turbine.	4	L2	3	1
VI.	(a) Derive an expression for the minimum speed for starting a centrifugal pump for flow to commence.	4	L3	4	1
	(b) There is a small colony in a village where poor people are living. The colony has a good compound wall but when roads were elevated, the houses got flooded due to water logging. The marriage of a girl in the colony is fixed and an NGO in which you are a member decided to pump out the water from this small colony continuously till the marriage function is over. The rate at which water sweeps in is at a rate of 1565 litres per second. The mean lift for pumping including losses is 6.1 m. You could hire a pump with the impeller dia meter 122 cm and impeller outer periphery area 6450 cm^2 . The vanes are setback at an angle of 26° at the outlet. The impeller can be rotated at 200 rpm. Considering overall efficiency, find the power of the prime mover (Engine or motor with suitable transmission system with a mechanical efficiency of 80%) to drive this pump. Also find the minimum speed to start pumping if the ratio of external to internal diameter of the impeller is 2.	8	L3	4	1
OR					
VII.	(a) Describe the principle and working of a reciprocating pump with a neat sketch. Define slip, percentage slip and negative slip of a reciprocating pump.	8	L3	4	1
	(b) A single – acting reciprocating pump, running at 60 r.p.m, delivers $0.01 \text{ m}^3/\text{s}$ of water. The diameter of the piston is 200 mm and stroke length 400 mm. Determine: (i) The theoretical discharge of the pump (ii) Coefficient of discharge and η_{vd} (iii) Slip and the percentage slip of the pump.	4	L3	4	1
VIII.	Explain the following with neat sketches Hydraulic press and Hydraulic accumulator.	12	L2	5	1
OR					
IX.	Explain the following with neat sketches Hydraulic intensifier and Gear pump.	12	L2	5	1

Blooms's Taxonomy Levels

L1 - 17%, L2 - 39%, L3 - 44%.